



Absolute Value of Complex Numbers

Algebra 2 Worksheet · Grade 10-12

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Learning Objectives

- Graph complex numbers in the complex plane
- Compute the absolute value (modulus) of a complex number using $|a+bi| = \sqrt{a^2 + b^2}$
- Compare distances of complex numbers from the origin

For each complex number, find its absolute value (modulus) and simplify, leaving radicals in exact form unless a decimal is requested.

1. Find the absolute value of the complex number.

$$|3 - 2i|$$

Answer: _____

2. Find the absolute value of the complex number.

$$|-2 + 5i|$$

Answer: _____

3. Find the absolute value of the complex number.

$$|-6i|$$

Answer: _____

4. Find the absolute value of the complex number.

$$|2 + 3i|$$

Answer: _____

5. Find the absolute value of the complex number.

$$|-4 + 2i|$$

Answer: _____

6. Find the absolute value of the complex number.

$$|7 - 24i|$$

Answer: _____

7. Find the absolute value of the complex number.

$$|-8 - 6i|$$

Answer: _____



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8. Which complex number is closer to the origin? Justify with absolute values.

$$|1 + 4i| \text{ vs. } |3 + 2i|$$

Answer: _____

9. Find the absolute value of the complex number.

$$|5 + 12i|$$

Answer: _____

10. Find the absolute value of the complex number.

$$\left| \frac{1}{2} - \frac{\sqrt{3}}{2}i \right|$$

Answer: _____





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ANSWER KEY & SOLUTIONS

Emphasize that $|a+bi|$ represents the distance from the origin to the point (a, b) in the complex plane; encourage students to sketch each number before computing.

Solutions

1. Find the absolute value of the complex number.

$$|3 - 2i|$$

- Identify $a = 3$ and $b = -2$.
- Apply the formula: square root of a squared plus b squared.
- Compute $9 + 4 = 13$.
- The absolute value is the square root of 13.

Answer: $\sqrt{13}$

2. Find the absolute value of the complex number.

$$|-2 + 5i|$$

- Identify $a = -2$ and $b = 5$.
- Square each: $(-2)^2 = 4$ and $5^2 = 25$.
- Add to get 29.
- The modulus is the square root of 29, approximately 5.39.

Answer: $\sqrt{29}$

3. Find the absolute value of the complex number.

$$|-6i|$$

- Rewrite as $0 + (-6)i$, so $a = 0$ and $b = -6$.
- Compute $0^2 + (-6)^2 = 36$.
- Take the square root of 36 to get 6.

Answer: 6

4. Find the absolute value of the complex number.

$$|2 + 3i|$$

- Identify $a = 2$ and $b = 3$.
- Square each: 4 and 9.
- Add to get 13.
- The absolute value is the square root of 13.

Answer: $\sqrt{13}$



5. Find the absolute value of the complex number.

$$|-4 + 2i|$$

→ Identify $a = -4$ and $b = 2$.

→ Square each: 16 and 4.

→ Add to get 20.

→ Simplify the square root of 20 as 2 times the square root of 5.

Answer: $2\sqrt{5}$

6. Find the absolute value of the complex number.

$$|7 - 24i|$$

→ Identify $a = 7$ and $b = -24$.

→ Square each: 49 and 576.

→ Add to get 625.

→ The square root of 625 is 25.

Answer: 25

7. Find the absolute value of the complex number.

$$|-8 - 6i|$$

→ Identify $a = -8$ and $b = -6$.

→ Square each: 64 and 36.

→ Add to get 100.

→ The square root of 100 is 10.

Answer: 10

8. Which complex number is closer to the origin? Justify with absolute values.

$$|1 + 4i| \text{ vs. } |3 + 2i|$$

→ Compute the first modulus: square root of $1 + 16 =$ square root of 17.

→ Compute the second modulus: square root of $9 + 4 =$ square root of 13.

→ Compare: square root of 13 is less than square root of 17.

→ Therefore $3 + 2i$ is closer to the origin.

Answer: $|3 + 2i| = \sqrt{13} < \sqrt{17} = |1 + 4i|$

9. Find the absolute value of the complex number.

$$|5 + 12i|$$

→ Identify $a = 5$ and $b = 12$.

→ Square each: 25 and 144.

→ Add to get 169.

→ The square root of 169 is 13.

Answer: 13



10. Find the absolute value of the complex number.

$$\left| \frac{1}{2} - \frac{\sqrt{3}}{2}i \right|$$

→ Identify $a = 1/2$ and $b = -\sqrt{3}/2$.

→ Square each: $1/4$ and $3/4$.

→ Add to get 1.

→ The square root of 1 is 1.

Answer: 1

